MIL-S-28600A(YD)
5 March 1976
SUPERSEDING
MIL-S-28600(YD)
21 January 1972

MILITARY SPECIFICATION

SORBENTS, FLOATING (FOR OIL SPILL REMOVAL)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers various sorbent materials used in conjunction with Navy oil spill removal.

* 1.2 Classification. The sorbents shall be of the following types, styles, and classes as specified (see 6.2):

Type I - Loose.
Type II - Flat (roll, sheet, or strip).
Type III - Enclosed.

Style 1 - Boom.
Style 2 - Pillow.

Class A - Reusable application.
Class B - Nonreusable application.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.
UU-S-48 - Sacks, Shipping, Paper.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

FSC 6850

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MILITARY

MIL-F-16884 - Fuel Oil, Diesel, Marine.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipping and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D828 - Tensile Breaking Strength of Paper and Paperboard.
D975 - Diesel Fuel Oils.
D1141 - Substitute Ocean Water.
D1596 - Shock Absorbing Characteristics of Cushioning Materials.
D1929 - Ignition Properties of Plastics.
D2859 - Flammability of Finished Textile Floor Covering Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. Sorbents shall be composed of materials that meet the requirements of this specification.

3.1.1 Solvent resistance. Materials may be coated or otherwise treated; however, they shall not be soluble or react chemically with oils at water temperatures below 37.8° Celsius (C).

3.1.2 Type III, style 1. The strength member (line) of type III, style 1 sorbent shall be made of nylon or polypropylene that complies with all requirements specified herein.

3.2 Dimensions.

3.2.1 Type I. Type I sorbent shall be furnished in loose pieces including chunks, fragments, granules, fibers, strips, or particles having a sieve analysis of not less than 90 ±2 percent retention when tested as specified in 4.4.1.

* 3.2.2 Type II. Type II sorbents shall be supplied in rolls, strips, or sheets. Unless otherwise specified (see 6.2), type II sorbent shall have a minimum thickness of 4.8 millimeters when tested as specified in 4.4.2.

* 3.2.3 Type III. Type III sorbents shall be comprised of either type I or type II sorbents completely enclosed in a run-resistant cover of open-mesh construction.

* 3.2.3.1 Type III, style 1. Type III, style 1 sorbent shall be furnished in the form of booms (sausage-shaped logs). Unless otherwise specified (see 6.2), booms shall have a minimum diameter of 12.7 centimeters (cm) a minimum length of 1.52 meters (m), and a maximum length of 3.04 m.

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* 3.2.3.2 Type III, style 2. Type III, style 2 sorbent shall be furnished in the form of pillows. Unless otherwise specified (see 6.2), the minimum thickness at the geometric center of the pillow shall be 10.2 cm when tested as specified in 4.4.2.

3.3 Performance characteristics.

3.3.1 Buoyancy. Less than 2 percent of the sorbent shall sink when tested as specified in 4.4.3.

* 3.3.2 Oil sorption ratio. The sorbent shall exhibit a minimum oil sorption ratio of 5.0 when tested as specified in 4.4.4.

* 3.3.3 Oil preference index. The sorbent shall have a minimum oil preference index of 20 when tested as specified in 4.4.5.

* 3.3.4 Storage safety.

* 3.3.4.1 Fire resistance. The sorbent in its unused dry state shall be rated as resistant to flammability when tested as specified in 4.4.6.

* 3.3.4.2 Ignition temperature. The sorbent shall have a self-ignition temperature greater than 148.9° C when tested as specified in 4.4.7.

* 3.3.5 Tensile strength (type II and type III only).

* 3.3.5.1 Type II. Type II sorbent shall have a minimum tensile strength in both principal directions of 90 kilopascals (kPa) in the dry condition and 62 kPa after being saturated with oil, when tested as specified in 4.4.8.

* 3.3.5.2 Type III, mesh cover. The mesh cover material enclosing type III sorbent shall have a minimum tensile strength in both principal directions of 11.3 kilograms (kg) per 2.54 cm width, in the dry condition and after being saturated with oil, when tested in accordance with 4.4.9.

* 3.3.5.3 Type III, style 1. The strength member and fastener of type III, style 1 sorbent shall have a minimum tensile strength of 272 kg when tested in accordance with 4.4.10.

* 3.3.6 Flotation (type III only). Type III sorbent shall not sink when tested as specified in 4.4.11.

* 3.3.7 Total oil pickup (type III only). Type III sorbent shall exhibit a minimum total oil pickup of 5.0 when tested as specified in 4.4.12.

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* 3.3.8 Reusability (class A only). Class A sorbent must be capable of being recycled a minimum of five times, while maintaining no less than 60 percent oil removal and retaining its original form, when tested as specified in 4.4.13.

3.3.9 Toxicity. Sorbents shall not be toxic to aquatic life and water fowl and shall have no adverse effect on the health of personnel when used for their intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as advisor to the procuring agency. A license shall be obtained from the appropriate state agency (if any) before sorbents are used in the waters of that state.

3.3.10 Pollution. Sorbents shall not of themselves or in combination with the oil, pollute waters or shorelines either chemically or physically when used for their intended purpose.

* 3.3.11 Disposal. When burned in an incinerator which meets local air quality requirements, sorbent combustion gases shall also meet these air quality standards.

* 3.3.12 Marking (type I only). A plainly printed label shall be attached to the type I sorbent container warning personnel against possible health hazards and recommending the use of suitable respiratory and eye protection against particulate matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Sampling. All sorbents of the same type, style, and class offered for delivery at one time shall be considered a lot for purposes of inspection. Random samples of the sorbents shall be selected from each lot in accordance with MIL-STD-105. The inspection level shall be S-3 and the Acceptable Quality Level 4.0 percent defective.

* 4.3 Examination. Each sample selected in accordance with 4.2 shall be examined for compliance with the requirements specified in Section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements.

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4.4 Tests. Sorbents selected in accordance with 4.2 shall be tested in accordance with the following methods. All tests involving oil shall use Marine diesel fuel oil conforming to MIL-F-16884 or diesel fuel oil grade No. 2-D conforming to ASTM D975.

4.4.1 Sieve analysis. An 0.5 liter sample of type I particles shall be shaken through a No. 100 U.S. standard sieve that conforms to RR-S-366, type I. The sample shall then be shaken for 15 minutes in a mechanically operated sieve shaker which imparts a rotary motion and tapping action of between 140 and 160 taps per minute. Alternatively, the sample may be shaken by hand for 20 minutes. The material retained in the screen shall be weighed and the percentage determined. Nonconformance to 3.2.1 shall constitute failure of this test.

4.4.2 Thickness, type II and type III, style 2 only. The thickness of type II and type III, style 2 sorbent shall be determined in accordance with Section 9(a) of ASTM D1596. Nonconformance to 3.2.2 or 3.2.3.2 shall constitute failure of this test.

* 4.4.3 Buoyancy. Approximately 5 gram (g) weighed to the nearest 0.01 g, of sorbent comprised of pieces having no dimension greater than 1.27 cm shall be placed in a wide mouth 0.95 liter jar half filled with synthetic sea water conforming to ASTM D1141. The jar and sorbent shall then be shaken on a reciprocating Eberback shaker table or equivalent at a frequency of 100 cycles per minute and a peak-to-peak amplitude of 2.54 cm for a period of 1 hour. Let the material stand for an additional 30 minutes without shaking. The floating sorbent shall then be removed with a ladle. The sunken portion of the sorbent shall also be removed and care taken to keep the two portions separated. Excess water shall be squeezed from both portions under the same conditions. Each portion shall be weighed separately, and the percent of the sunken sorbent shall be calculated. This test shall be repeated for five samples and the resulting percentage measures of sunken sorbent averaged for the single value required in 3.3.1. Nonconformance to 3.3.1 shall constitute failure of this test.

* 4.4.4 Oil sorption ratio. A sorbent sample of approximately 5 g, weighed to the nearest 0.01 g shall be prepared by cutting squares having edge dimensions of approximately 1.27 cm. Fill an open top pan, of surface area equal to 150 percent of the total surface area of the squares in the 5 g sample as they would appear when floating on an oil slick, with oil to a depth equal to twice the thickness of the sorbent squares. Place all of the squares in the 5 g sample on top of the oil and allow to stand for 15 minutes. At the end of this period carefully remove the saturated sorbent using a 90 percent open mesh screen tray

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lifted vertically. Allow the sorbent to drain for 5 minutes, weigh the saturated sorbent and calculate the oil sorption ratio as follows:

$$\text{Oil sorption ratio} = \frac{\text{Saturated wt. - dry wt.}}{\text{Dry wt.}}$$

The above procedure shall be repeated for five sorbent samples and the resulting oil sorption ratios averaged and reported as the oil sorption ratio for the particular sorbent. Nonconformance to 3.3.2 shall constitute failure of this test.

* 4.4.5 Oil preference index. The oil preference index is a coefficient which indicates the relative ability for sorbing oil as opposed to water. The average water sorption ratio shall be determined in accordance with 4.4.4 except that synthetic ocean water prepared in accordance with ASTM D1141 shall be used instead of oil. The oil preference index is calculated as follows:

$$\text{Oil preference index} = \frac{\text{Average oil sorption ratio (five samples)}}{\text{Average water sorption ratio (five samples)}}$$

Nonconformance to 3.3.3 shall constitute failure of this test.

* 4.4.6 Fire resistance. Fire resistance of the sorbent shall be determined in accordance with ASTM D2859. Nonconformance to 3.3.4.1 shall constitute failure of this test.

* 4.4.7 Ignition temperature. The self-ignition temperature shall be determined in accordance with ASTM D1929. Nonconformance to 3.3.4.2 shall constitute failure of this test.

* 4.4.8 Tensile strength, type II. The tensile strength of type II sorbent shall be determined in accordance with ASTM D828. Sample specimens not less than 1.27 cm and not more than 5.08 cm wide shall be tested on a motor-driven tensile breaking strength testing apparatus. Five specimens shall be tested in both principal directions in the dry condition and after being submerged in oil for 1 hour. Results shall be reported separately for each principal direction for the dry and oil saturated conditions. The tensile strength shall be the average of the results obtained from the five specimens and shall be reported to the nearest kPa. Nonconformance to 3.3.5.1 shall constitute failure of this test.

4.4.9 Tensile strength, type III, mesh cover. The tensile strength of the mesh cover enclosing type III sorbent shall be determined in accordance with ASTM D828. Sample specimens not less than 1.27 cm and not more than 5.08 cm wide shall be tested on a motor-driven tensile

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breaking strength testing apparatus. Five specimens shall be tested in both principal directions in the dry condition and after being submerged in oil for 1 hour. Results shall be reported separately for each principal direction for the dry and oil saturated conditions. The tensile strength shall be the average of the results obtained for the five specimens and shall be reported to the nearest 0.1 kg. Nonconformance to 3.3.5.2 shall constitute failure of this test.

* 4.4.10 Tensile strength, type III, style 1. The fastener and strength member of type III sorbent shall be tested on an "Instron Tester" or equivalent. The tensile strength shall be the average of the results obtained from five specimens and shall be reported to the nearest kg. Nonconformance to 3.3.5.3 shall constitute failure of this test.

* 4.4.11 Flotation (type III). A sample of the assembled type III sorbent, complete with enclosing mesh, strength line, and fastener shall be placed in synthetic ocean water conforming to ASTM D1141 for a period of 1 hour. Nonconformance to 3.3.6 shall constitute failure of this test.

* 4.4.12 Total oil pickup, type III. Prepare an 0.61 m test section of the type III sorbent, having the same cross sectional area, sorbent particle size, and packing density as in the delivered product. Record the total weight of the dry sorbent contained inside the mesh enclosure. Place the 0.61 m test section in a pan filled with oil to a depth greater than that of the test section diameter. Measure the height of the sorbent type III above the free surface of the oil and record. Continue measuring until a constant height has been achieved. Drain test section horizontally for 1 minute and weigh. Subtract from this weight the weight of the mesh enclosure, strength lines, and fasteners, if present, to obtain the weight of the oil and sorbent alone. Calculate total oil pickup as follows:

$$\text{Total oil pickup} = \frac{\text{Wt. of oil and sorbent} - \text{Wt. dry sorbent}}{\text{Wt. dry sorbent}}$$

Nonconformance to 3.3.7 shall constitute failure of this test.

* 4.4.13 Reusability. An 8.20 liter volume of sorbent shall be placed in a bucket-type janitorial mop squeezing assembly. Known weight of oil shall be poured into the sorbent until it is completely saturated. Any oil which has drained from the sorbent during this process shall be collected and weighed. This weight shall be subtracted from the total weight of the oil poured into the sorbent, and the result recorded as the oil weight at saturation. The saturated sorbent shall then be squeezed in the mop assembly at a pressure of 69-90 kPa for 30 seconds, or until

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oil stops dripping from the sorbent, whichever occurs first. The oil squeezed from the sorbent shall be collected, weighed, and recorded as oil weight recovered. The percent oil removal shall then be calculated as follows:

$$\text{Percent oil removal} = \frac{\text{Oil weight recovered}}{\text{Oil weight at saturation}} \times 100$$

Repeat this procedure fives times. Any serious degradation in sorbent structural integrity shall be cause for rejection. Nonconformance to 3.3.8 shall constitute failure of this test.

4.5 Packaging inspection. The preservation-packaging, packing, and marking shall be inspected to determine conformance to the requirements of Section 5 of this specification.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

* 5.1.1.1 Type I. Type I sorbent shall be packed in a shipping sack conforming to UU-S-48, No. 8-8X, not to exceed 22.7 kg.

* 5.1.1.2 Type II and III. Type II and III sorbent shall be packed in fiberboard boxes conforming to PPP-B-636, type SF, class weather resistant, grade V2s, or when loads exceed the limits of PPP-B-636, in containers conforming to PPP-B-640, class weather resistant.

5.1.2 Level B.

* 5.1.2.1 Type I. Type I sorbent shall be packed in a shipping sack conforming to UU-S-48, No. 4-4X, not to exceed 22.7 kg.

* 5.1.2.2 Type II and III. Type II and III sorbent shall be packed in fiberboard boxes conforming to PPP-B-636, type CF, class weather resistant, grade V3c, or when loads exceed the limits of PPP-B-636, in containers conforming to PPP-B-640, class domestic.

5.1.3 Level C. Sorbent shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. In addition to any special marking required by the contract, the sorbent materials shall be marked in accordance with MIL-STD-129.

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* 5.3 Palletization and containerization. When specified (see 6.2), sorbent packed in accordance with 5.1 shall be palletized in accordance with MIL-STD-147, or overpacked in boxes conforming to PPP-B-601, overseas type; or PPP-B-621, class 2.

6. NOTES

* 6.1 Intended use. Sorbents covered by this specification are intended to be applied over oil spills for the purpose of sorbing and subsequently harvesting the oil, either manually or mechanically.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and class of sorbent required (see 1.2).
(c) Dimensions of type II sorbent (see 3.2.2).
(d) Dimensions of type III, style 1 sorbent (see 3.2.3.1).
(e) Dimensions of type III, style 2 sorbent (see 3.2.3.2).
(f) Level of packing required (see 5.1).
(g) When palletization or containerization is required (see 5.3).

* 6.3 Metric conversion. The following table is included for conversion from metric measures:

Paragraph	Metric	English
3.1.1	37.8° Celsius	100° Fahrenheit
3.2.2	4.8 millimeters	3/16 inch
3.2.3.1	12.7 centimeters	5 inches
3.2.3.1	1.52 meters	5 feet
3.2.3.1	3.04 meters	10 feet
3.2.3.2	10.2 centimeters	4 inches
3.3.4.2	148.9° Celsius	300° Fahrenheit
3.3.5.1	90 kilopascals	13 psi
3.3.5.1	62 kilopascals	7 psi
3.3.5.2	11.3 kilograms	25 pounds
3.3.5.2, 4.4.3	2.54 centimeters	1 inch
3.3.5.3	272 kilograms	600 pounds
4.4.3	0.95 liter	1 quart
4.4.3, 4.4.4, 4.4.8	1.27 centimeters	1/2 inch
4.4.8	5.08 centimeters	2 inches
4.4.11, 4.4.12	0.61 meter	2 feet
4.4.13	8.20 liter	500 cubic inches
5.1.1.1, 5.1.2.1	22.7 kilograms	50 pounds

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* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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